DATA ANALYSIS PROJECT

Is South Carolina Heating Up?

Three angles on 60 years of daily weather data (1966–2025) from NOAA

Student: [Your Name]
Course: [Course name and number]
Instructor: [Instructor name]
Date: [Submission date]
Code & data: [GitHub repository link]
Video: [YouTube presentation link]

Summary

This project downloads 60 years of daily weather records for 10 long-running South Carolina weather stations from NOAA's National Centers for Environmental Information (NCEI) and looks at climate change in the state from three angles:

- **Angle 1 – Where?** Which city has warmed the most? South Carolina as a whole warmed **+0.49 °F per decade** (about **+2.9 °F** since 1966). **Columbia** warmed fastest (+0.58 °F/decade), but it is statistically tied with Greenville-Spartanburg, Beaufort and Charleston Airport. Every trusted station warmed significantly.
- **Angle 2 – When?** Day or night, and which season? Nights are warming faster than afternoons (+0.55 vs +0.43 °F/decade), and the gap is biggest in the two most urban stations, downtown Charleston and Columbia. **Winter** is warming 1.7× faster than the other seasons.
- **Angle 3 – What do people feel?** Extreme days. Warm nights (low ≥ 75 °F) are up **47%**, freezing nights are down **18%**, and the freeze-free growing season is about **3 weeks longer** than in the 1960s. Heavy rain, on the other hand, shows **no significant trend** – big storm years happen in every decade.

Along the way, a simple neighbour check showed that one station (Sumter) has a broken temperature record, so it was left out of the statewide numbers. All code is in Python (pandas, NumPy, SciPy) and every figure is made with Matplotlib. Running one command (python run_all.py) rebuilds every number and figure in this report.

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1. Introduction and motivation

Charleston, South Carolina is the example city in the assignment, and it is a good place to start: it floods on sunny days at high tide, it was hit by the October 2015 "1000-year" flood, and summers there are famously hot and sticky. We hear a lot about climate change at the global level, but people experience the weather where they live. So the question for this project is simple: **what has actually changed in South Carolina's weather over the last 60 years, and where, when and how do people feel it?**

This is interesting for three reasons:

- **It is local and checkable.** Instead of a global average, we use the real thermometer and rain-gauge readings from airports and universities that people in the state know.
- **It matters for decisions.** Farmers care about freeze dates, cities care about heat waves and energy demand, hospitals care about hot nights, and engineers care about heavy rain.
- **It is a good data-science exercise.** Raw weather data is messy: values are missing, instruments change and stations move. Getting an honest answer means cleaning the data, checking it and testing whether a change is bigger than natural year-to-year noise.

2. Task 1 – Downloading the data

2.1 Data source

All data comes from NOAA NCEI's **Global Historical Climatology Network – Daily (GHCN-Daily)**. This is the same database behind the NOAA "Past Weather" page for Charleston that the assignment links to. Each station has one CSV file that can be downloaded directly, with no account or API key:

```
https://www.ncei.noaa.gov/data/global-historical-climatology-network-
daily/access/<STATION_ID>.csv
```

Each row is one day at one station. The columns used in this project are the daily high temperature (TMAX), daily low temperature (TMIN) and daily rain (PRCP), plus NOAA's quality flags for each value.

2.2 Choosing the stations

South Carolina has dozens of weather stations, but most have short or broken records. To pick stations I downloaded NOAA's station list (ghcnd-stations.txt) and inventory (ghcnd-inventory.txt), filtered them to South Carolina stations whose records start before 1955 and run to 2025, then downloaded 18 candidates and counted how many complete years each one had. A year counts as complete if it has at least **330 days** of data (about 90%).

I kept **10 stations** that are complete for almost every year from **1966 to 2025** and that cover the state's three well-known regions – the **Coast**, the **Midlands** and the **Upstate**. Charleston has two stations (the airport and downtown), which turned out to be very useful for seeing the effect of the city itself.

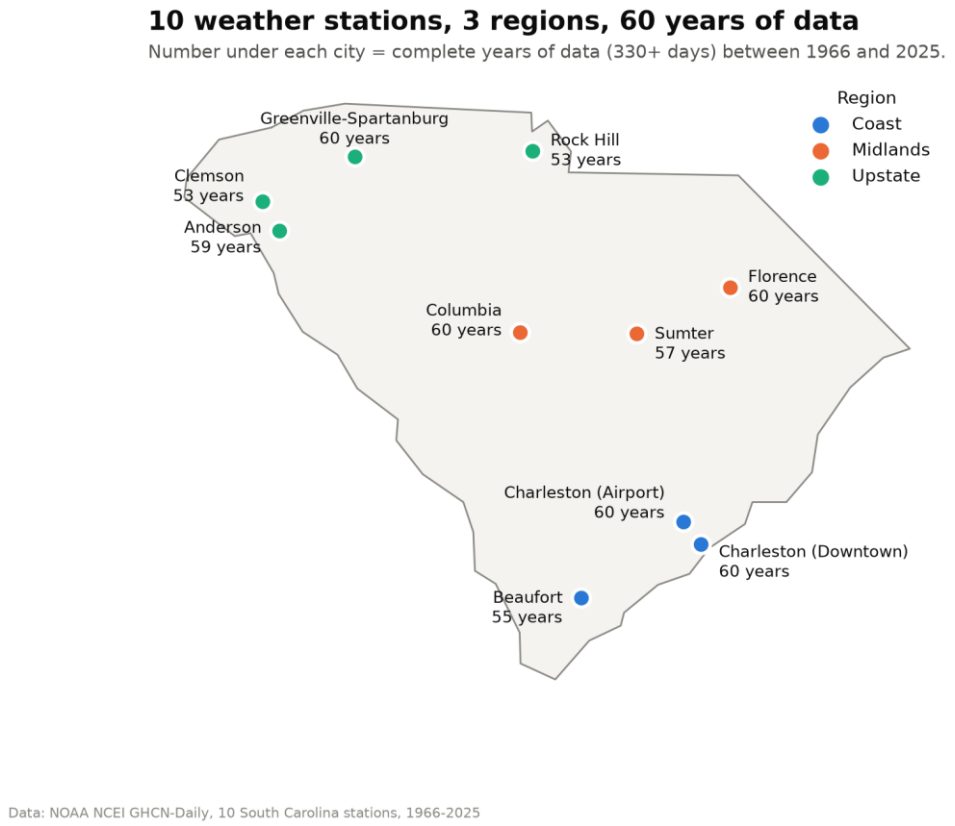


Figure 1. The 10 stations and the number of complete years (≥ 330 days) each one has between 1966 and 2025.

Table 1. The 10 stations used, with data completeness after cleaning.

City	Region	NOAA station ID	Temp. years	Rain years	Missing temp. days
Charleston (Airport)	Coast	USW00013880	60	60	0.0%
Charleston (Downtown)	Coast	USW00013782	60	60	0.2%
Beaufort	Coast	USW00093831	55	55	1.1%
Columbia	Midlands	USW00013883	60	60	0.0%
Florence	Midlands	USW00013744	60	60	0.1%
Sumter	Midlands	USC00388440	57	59	2.6%
Greenville-Spartanburg	Upstate	USW00003870	60	60	0.0%
Anderson	Upstate	USW00093846	59	59	0.1%
Clemson	Upstate	USC00381770	53	60	3.3%
Rock Hill	Upstate	USC00389350	53	58	3.3%

2.3 The download script

The script **src/download_data.py** loops over the 10 station IDs (stored in src/config.py), downloads each CSV with Python's built-in urllib, keeps only the date, station information, PRCP/TMAX/TMIN and their quality-flag columns, and saves a trimmed copy in **data/raw/**. The full NOAA files have about 100

columns (wind, snow depth and so on), so trimming makes the raw data about 27 MB instead of more than 100 MB. The trimmed files are committed to the repository, so the project runs without internet; python run_all.py --download fetches fresh data.

3. Task 2 – Cleaning and analysing the data

3.1 Cleaning (src/clean_data.py)

1. **Unit conversion.** NOAA stores temperature in tenths of a degree Celsius and rain in tenths of a millimetre. Everything is converted to °F and inches, because those are the units people in South Carolina use.
2. **Quality flags.** Every value has a flag field ("M,Q,S,time"). If NOAA's quality-control flag Q is set, the value failed one of NOAA's checks, so it is removed. This removed, for example, 685 daily lows at Clemson and 533 daily highs at Rock Hill.
3. **Impossible days.** Days where the high is lower than the low are removed (none were left after step 2).
4. **Complete years only.** Yearly numbers are only calculated for years with at least 330 valid days; otherwise a missing summer month could make a year look cooler than it was.
5. **Yearly table.** For each station and year the script calculates the average temperature, average high and low, the number of hot days, warm nights and freeze days, the dates of the last spring and first fall freeze, the rain total, number of heavy-rain days, wettest day and the longest dry spell. This table (data/processed/annual_metrics.csv) is what the three angles use.

3.2 A data-quality check: the neighbour test (src/quality_check.py)

Weather stations get moved, re-sited and fitted with new sensors. Such changes can make a station look as if it suddenly warmed or cooled when the climate did not. Because stations in the same state rise and fall together from year to year, I compared each station with the average of the other nine:

```
difference = station's temperature anomaly − average anomaly of the other 9 stations
```

For a healthy station the 5-year average of this difference stays close to zero (all of them stayed within about 2 °F). **Sumter** was the exception: from about 2005 to 2015 its daily highs read up to **5 °F colder** than its neighbours, then recovered. That is not real weather – Columbia, 40 miles away, shows nothing like it – so Sumter was flagged as "suspect" and left out of all statewide temperature averages. It is still shown, in grey, on the charts. Its rain gauge was not affected, so its rain data is still used.

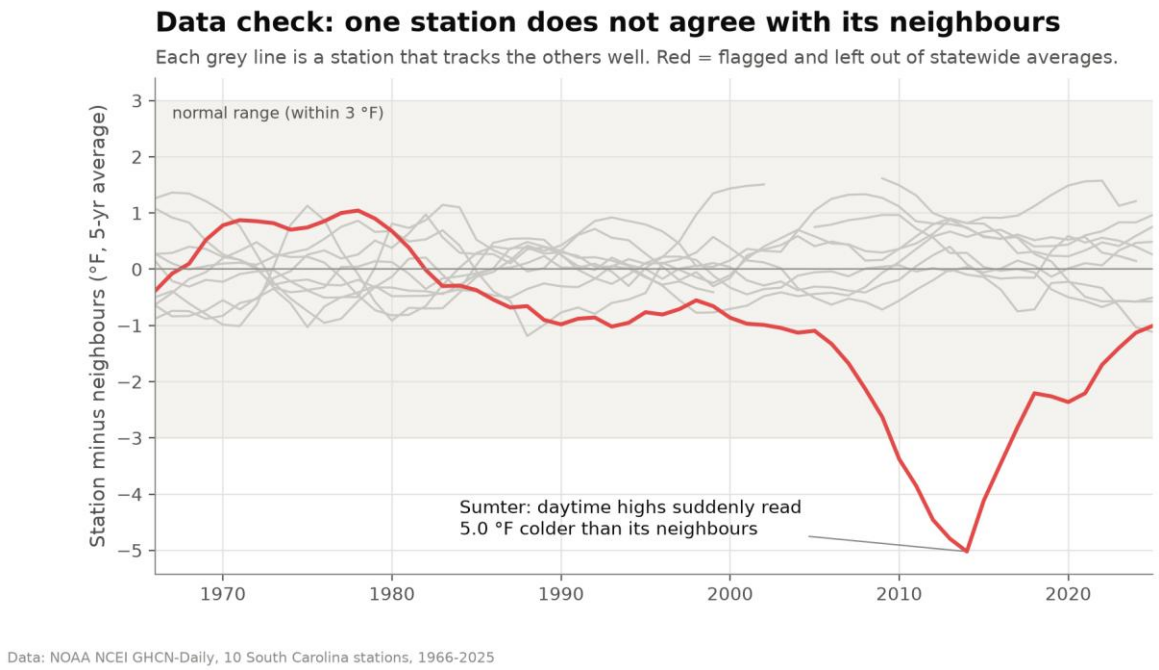


Figure 2. Neighbour check. Each grey line is a station minus the average of the others (5-year average of daily highs). Sumter (red) drifts far outside the normal range.

3.3 Statistics used

- **Linear trend.** For each series I fit a straight line through the 60 yearly values with least squares (scipy.stats.linregress). The slope × 10 gives the change **per decade**.
- **95% confidence interval and p-value.** The p-value answers "how likely is a slope this big if there were really no trend and the data were just random year-to-year noise?". A trend with $p < 0.05$ is called significant and marked with stars (*p<0.05, p<0.01,* ** p<0.001).
- **Then vs. now.** I also compare the average of 1966–1995 with 1996–2025. These are two 30-year periods, the same length NOAA uses for its climate "normals", so the comparison is easy to explain.
- **Anomalies.** For statewide averages each station is first expressed as a difference from its own 1966–1995 average. This way a cool mountain station and a warm coastal station can be averaged fairly, and one missing year at one station doesn't bias the result.

4. The three angles and why they make sense

Climate change is often described with one number – "the average temperature went up". But one number can't tell a city planner where to act, a doctor when people are at risk or a farmer when to plant. I chose three angles that each answer the question the previous one leaves open:

Table 2. The three angles, the question each answers and why it matters.

Angle	Question	What it measures	Why it makes sense
1. WHERE	Which South Carolina city has warmed the most?	Trend of each city's yearly average temperature (space)	This is the example question in the assignment. It tells us whether warming is local or statewide and gives a ranking that local governments can use.
2. WHEN	Is it warming more by day or by night? In which season?	Separate trends for daily highs vs. lows, and for winter/spring/summer/fall (time)	The yearly average hides *how* it warms. Night vs. day points to causes such as city heat and humidity; season matters for crops, pests and energy use.
3. WHAT PEOPLE FEEL	Are extreme days becoming more common?	Counts of hot days, warm nights, freezes, growing season, heavy rain (extremes)	People feel thresholds, not averages: a 95 °F afternoon, a night that never cools, a frost or a flood. This angle also tests rain, a second variable.

Together the three angles move from **the average** (Angle 1) to **its structure in time** (Angle 2) to **the extremes** (Angle 3). They look at space, time and impacts, they use the same cleaned data and the same statistics, and they also guard against fooling ourselves: if the average warmed but extremes did not change, or if only one station showed warming, the story would be very different.

5. Angle 1 – Where is South Carolina warming the most?

5.1 Method

For every station I took the yearly average temperature ((daily high + daily low) / 2, averaged over all days of the year) and fitted a straight-line trend from 1966 to 2025. The statewide line is the average anomaly of the 9 trusted stations.

5.2 Results

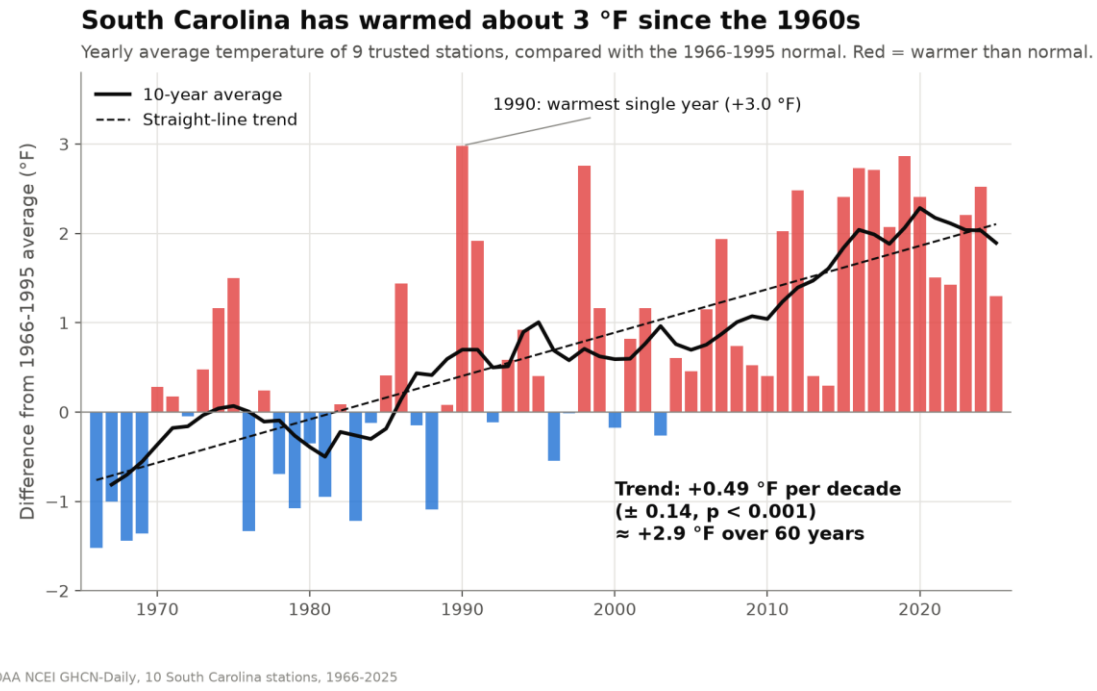


Figure 3. Statewide yearly temperature compared with the 1966–1995 normal. Since 2010 almost every year has been warmer than normal.

South Carolina has warmed by **+0.49 °F per decade** (95% CI ± 0.14, p < 0.001), which adds up to about **+2.9 °F over 60 years**. The last 30 years were on average about 1.4 °F warmer than the 30 years before. The warmest single year was 1990, but **8 of the 10 warmest years came after 2010** (2019, 2016, 2017, 2024, 2012, 2020, 2015 and 2023).

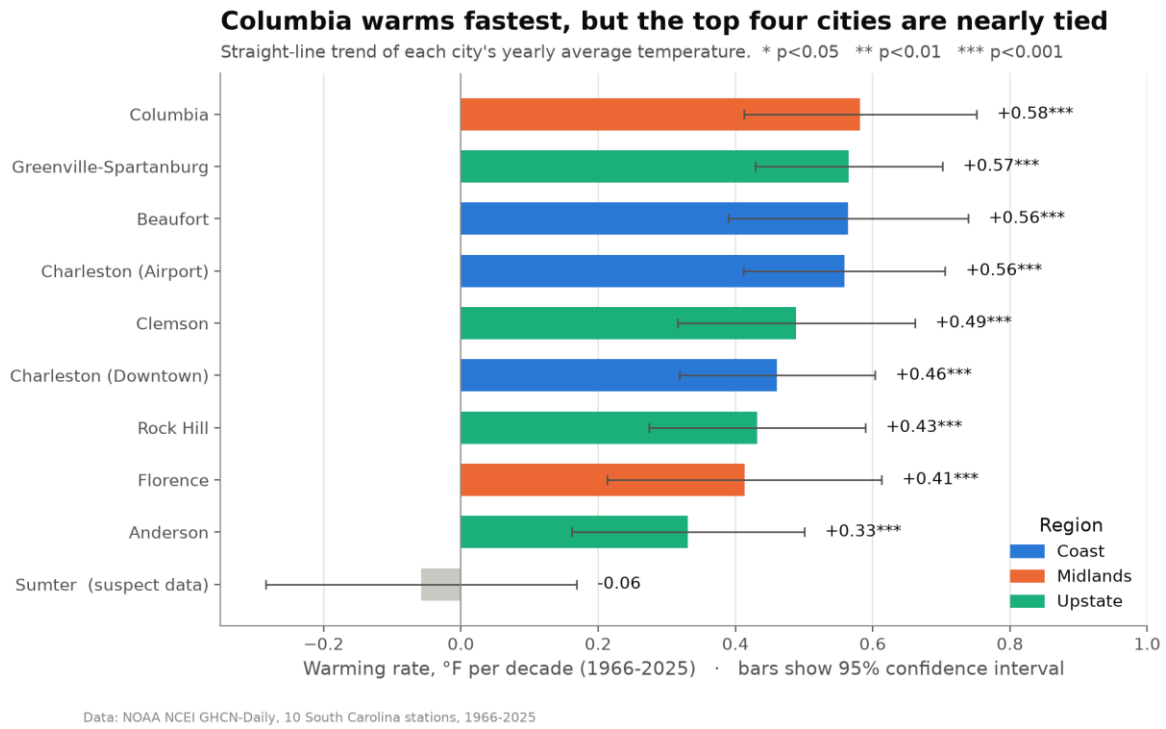


Figure 4. Warming rate of each city with 95% confidence intervals. Colours show the region; Sumter (grey) failed the data check.

Table 3. Warming rate of each city, 1966–2025 (ranked).

Rank	City	Region	°F per decade	95% CI	Sig.	1966–95 → 1996–2025 (°F)
1	Columbia	Midlands	+0.58	± 0.17	***	63.3 → 64.9
2	Greenville-Spartanburg	Upstate	+0.56	± 0.14	***	60.0 → 61.8
3	Beaufort	Coast	+0.56	± 0.17	***	66.3 → 67.7
4	Charleston (Airport)	Coast	+0.56	± 0.15	***	65.6 → 66.9
5	Clemson	Upstate	+0.49	± 0.17	***	60.3 → 62.2
6	Charleston (Downtown)	Coast	+0.46	± 0.14	***	66.4 → 67.8
7	Rock Hill	Upstate	+0.43	± 0.16	***	61.3 → 62.7
8	Florence	Midlands	+0.41	± 0.20	***	63.7 → 64.6
9	Anderson	Upstate	+0.33	± 0.17	***	61.6 → 62.4
10	Sumter (suspect)	Midlands	-0.06	± 0.23	n.s.	63.2 → 63.0

Answer to Angle 1: Columbia has warmed the fastest, at **+0.58 °F per decade** – about +3.5 °F since 1966. However, the top four (Columbia, Greenville-Spartanburg, Beaufort, Charleston (Airport)) are within 0.03 °F/decade of each other and their confidence intervals overlap almost completely, so it is fairer to say these four are statistically tied. The slowest-warming trusted station is Anderson (+0.33). If

we instead compare the two 30-year averages, Clemson (+1.9 °F) and Greenville-Spartanburg (+1.8 °F) show the biggest jump, so the Upstate and Columbia stand out either way.

A second, maybe more important, finding is that **all 9 trusted stations warmed significantly** ($p < 0.001$ for every one). The region averages are very close – Coast +0.53, Midlands +0.50, Upstate +0.45 °F/decade – so warming is a statewide change, not a local one.

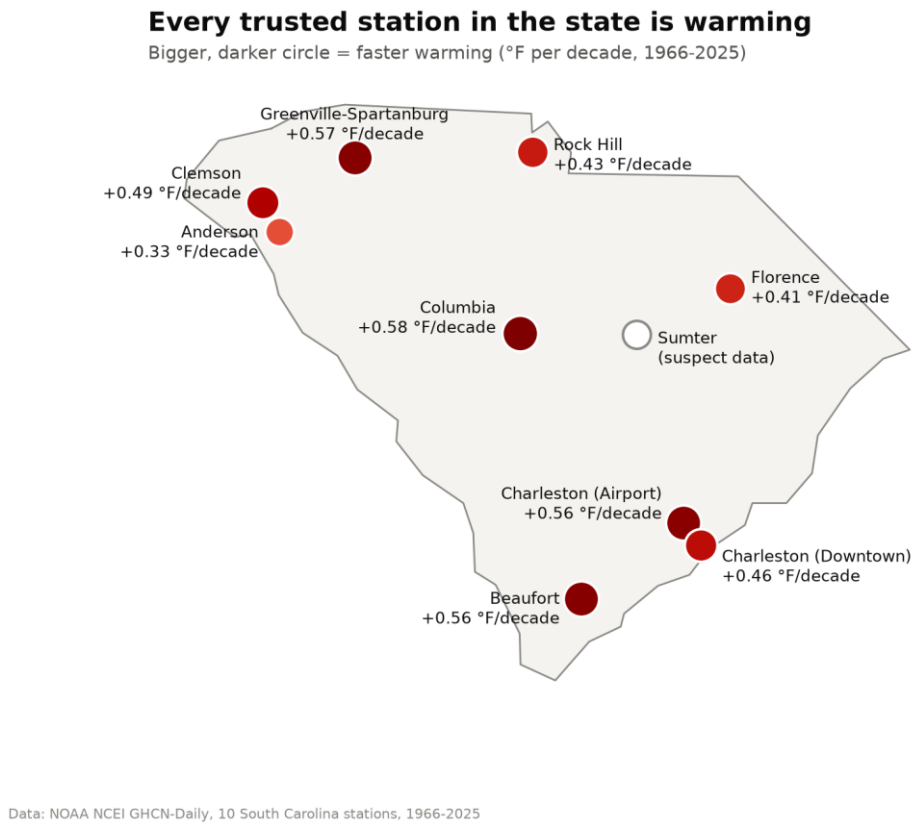


Figure 5. The same warming rates on a map. Bigger and darker circles mean faster warming.

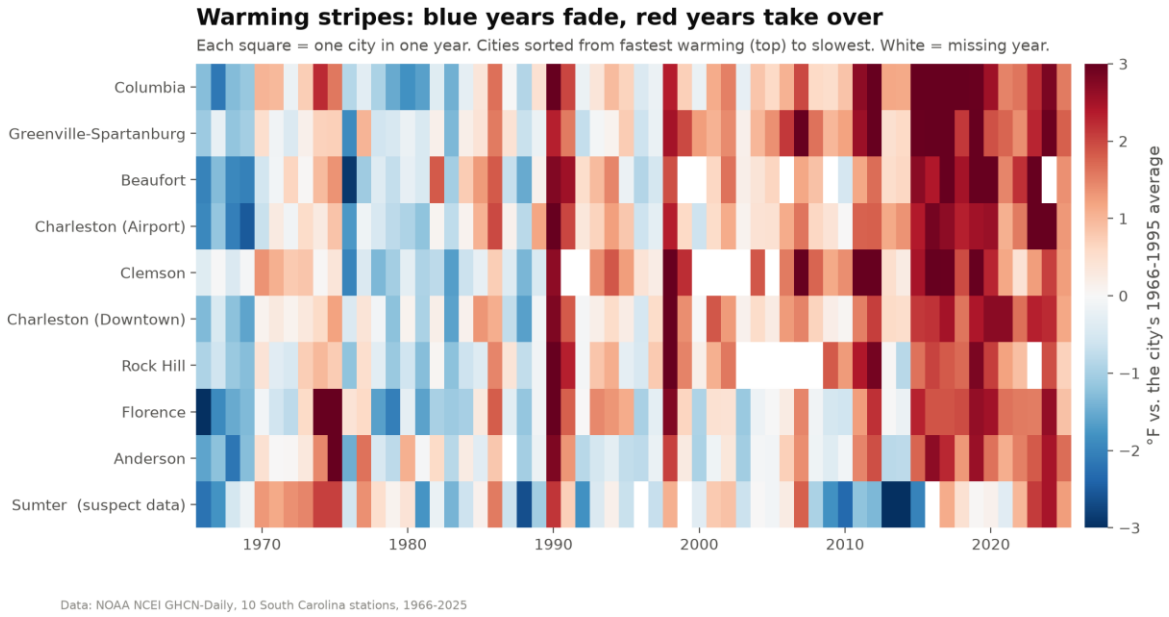


Figure 6. "Warming stripes": every city (row) in every year (column), coloured by how much warmer (red) or cooler (blue) it was than that city's 1966–1995 average.

The warming-stripes chart (Figure 6) shows the same story in one picture without any statistics: the left side is mostly blue, the right side is mostly red, and the change happens at all stations at the same time. It also makes Sumter's problem visible – it is the only row that turns dark blue around 2013–2014.

6. Angle 2 – When: day or night, and which season?

6.1 Method

Instead of the daily average, I fit separate trends for the **daily high** (the afternoon temperature, TMAX) and the **daily low** (the night-time / early-morning temperature, TMIN). Then I split each year into the four meteorological seasons (winter = December–February, with December counted in the next year's winter) and fit a statewide trend for each season. A season is only used if it has at least 80 days of data.

6.2 Day vs. night

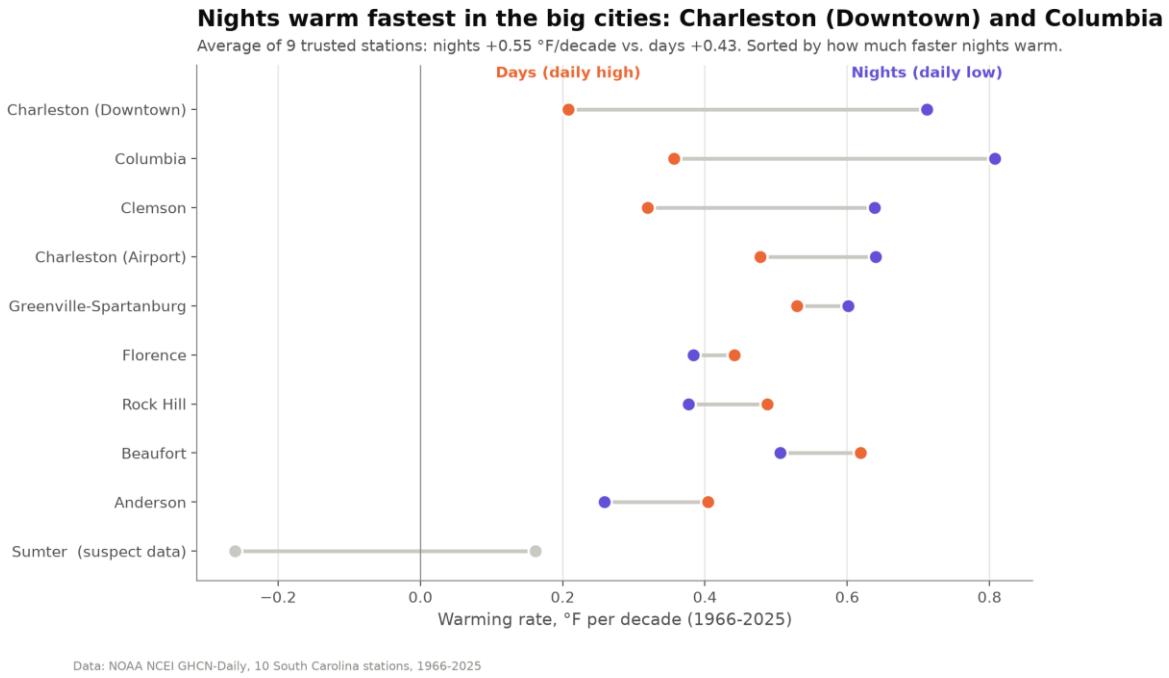


Figure 7. Warming of the daily high (orange) and the daily low (purple) at each station, sorted by how much faster nights warm than days.

Averaged over the 9 trusted stations, nights warmed **+0.55 °F/decade** and afternoons **+0.43 °F/decade**, so nights are warming about 28% faster. But the most interesting part is *where* the gap is largest: in **downtown Charleston** nights warmed 0.71 °F/decade but days only 0.21 (more than 3× faster), and in **Columbia** 0.81 vs 0.36 °F/decade. These are the two most built-up locations in the study. Smaller and more rural stations (Anderson, Rock Hill, Florence, Beaufort) warm about equally or slightly more by day.

This matches the **urban heat island** effect: concrete, asphalt and buildings soak up heat during the day and release it at night, so cities stay warm after sunset. More humid air also traps heat at night. In other words, part of the warming at city stations is extra warming caused by the city around the thermometer, on top of the regional climate change seen at every station.

6.3 Which season?

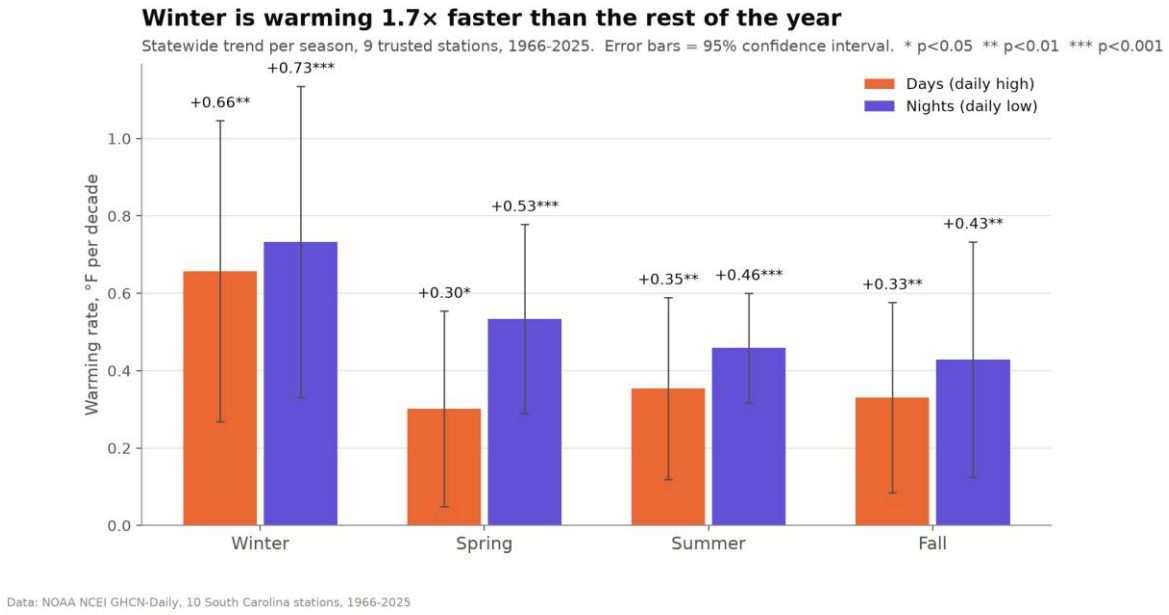


Figure 8. Statewide warming rate for each season, for daily highs and lows, with 95% confidence intervals.

Table 4. Statewide warming by season, 1966–2025 (°F per decade).

Season	Daily high	Sig.	Daily low	Sig.
Winter	+0.66 ± 0.39	**	+0.73 ± 0.40	**
Spring	+0.30 ± 0.25	*	+0.53 ± 0.24	***
Summer	+0.35 ± 0.23	**	+0.46 ± 0.14	***
Fall	+0.33 ± 0.25	**	+0.43 ± 0.30	**

Winter is warming fastest – about +0.66 °F/decade for highs and +0.73 for lows, **1.7× faster** than the average of the other three seasons. In the other seasons the night-time lows clearly warm faster than the afternoon highs (for example, spring nights +0.53 vs spring afternoons +0.30). Winter temperatures also vary a lot from year to year (a cold January versus a mild one), so the winter confidence intervals are wide; the winter result is significant, but the exact factor of 1.7 is less certain than the others.

Answer to Angle 2: South Carolina is warming most in **winter** and at **night**, and the night-time warming is strongest in the cities. This is important because the averages in Angle 1 hide it: a +2.9 °F average is mostly milder winters and warmer nights, not uniformly hotter afternoons.

7. Angle 3 – What do people feel? Extreme days

7.1 Method

For each station and year I counted days that cross thresholds people actually notice: **hot days** (high ≥ 95 °F), **warm nights** (low ≥ 75 °F, when the body and air-conditioners get no break), **freeze days** (low ≤

32 °F), the **freeze-free season** (days between the last spring freeze and the first fall freeze – roughly the growing season), **heavy-rain days** (≥ 2 inches in one day – enough to cause flash flooding), plus the yearly rain total, the wettest day of the year and the longest dry spell. These are similar to the internationally used ETCCDI climate-extreme indices. I compare 1966–1995 with 1996–2025 and test the 60-year trend.

7.2 Heat and cold

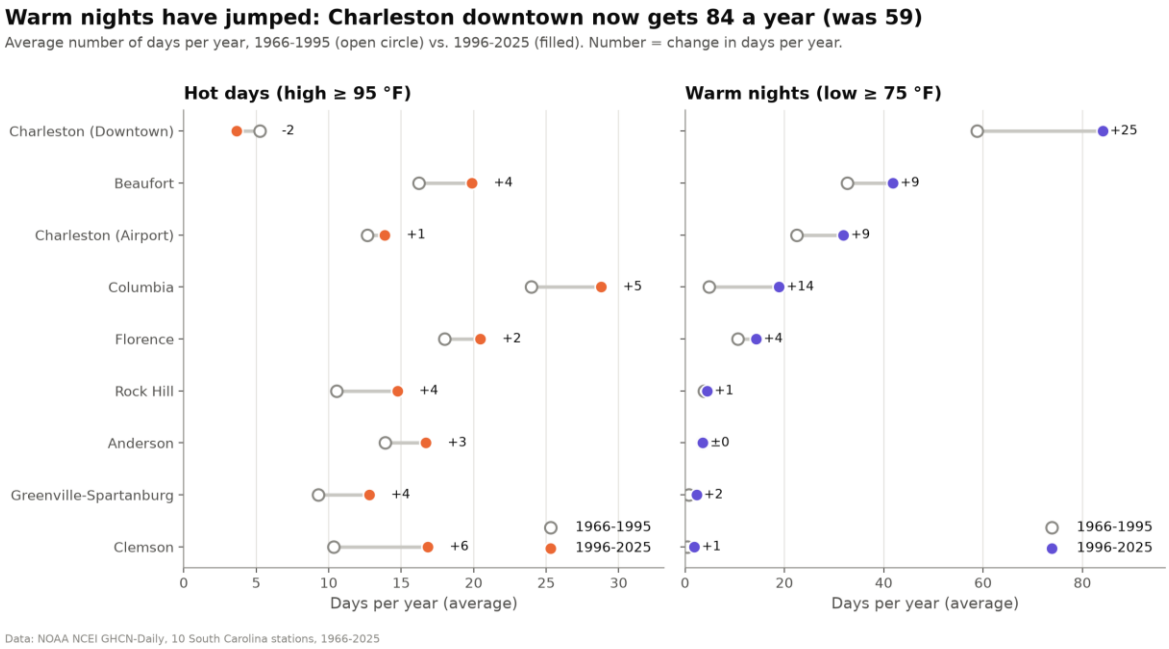


Figure 9. Average number of hot days and warm nights per year, 1966–1995 (open circle) vs. 1996–2025 (filled circle). The number is the change in days per year.

Warm nights changed the most: statewide they went from 15.6 to 22.9 per year (**+47%**, $p < 0.001$). Downtown Charleston now averages **84** nights a year that never drop below 75 °F, up from 59 – almost a whole extra month. Columbia went from about 5 to 19 per year (nearly 4×). **Hot days** rose less, from 13.4 to 16.2 per year (+21%, $p = 0.046$, just significant). Interestingly downtown Charleston, cooled by the sea breeze, has *fewer* hot afternoons than before – again showing that the change there is about the nights.

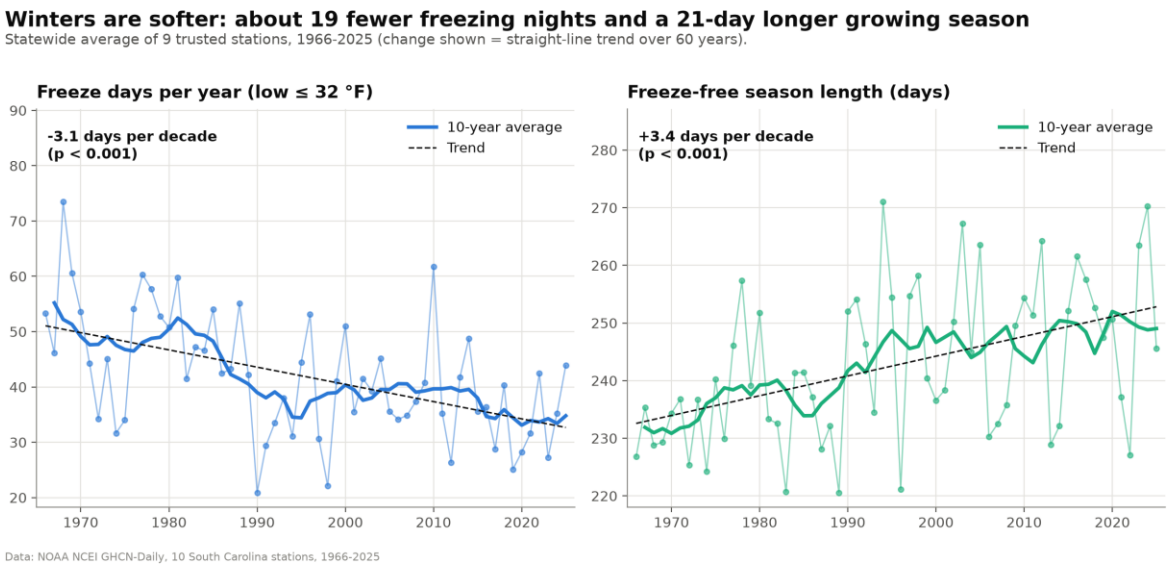


Figure 10. Statewide freeze days per year (left) and length of the freeze-free season (right), with 10-year averages and straight-line trends.

At the cold end, **freeze days** fell from 46.0 to 37.7 per year (-18%; trend −3.1 days per decade, about 19 fewer freezing nights over 60 years). The **freeze-free season** grew by 3.4 days per decade – about **3 weeks longer** than in the 1960s. Clemson lost the most freezing nights (70 → 54 per year) and downtown Charleston's freeze-free season is now about 324 days long.

7.3 Rain

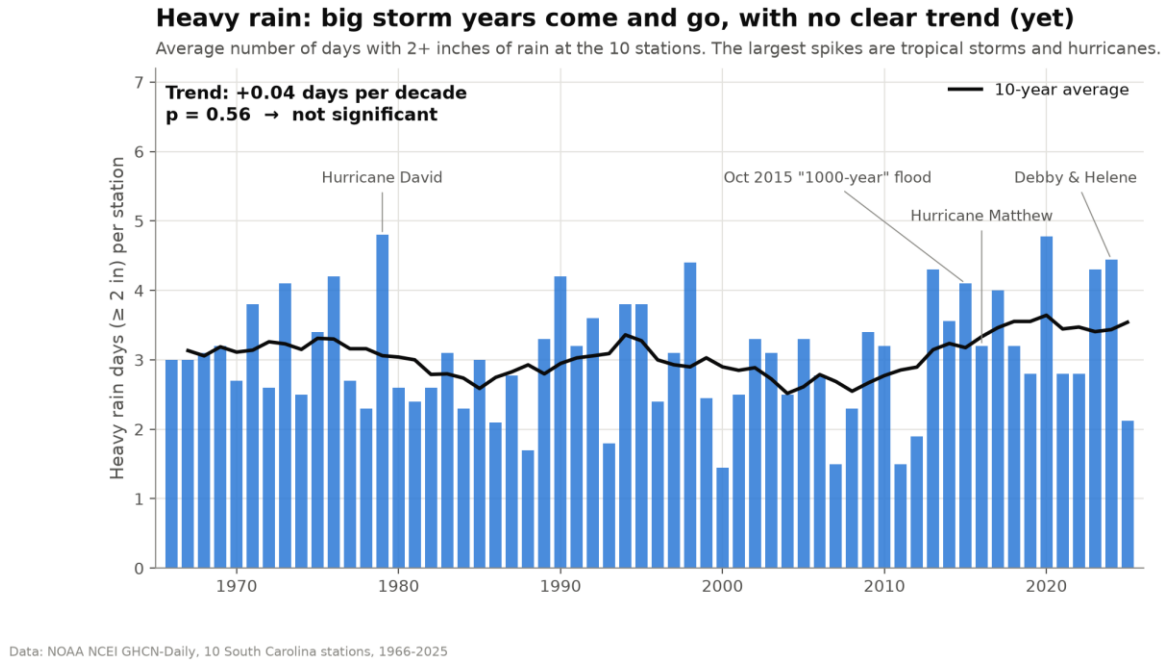


Figure 11. Average number of days with 2 inches or more of rain per station, with famous storm years labelled.

Rain tells a different story. The number of heavy-rain days has **no significant trend** (p = 0.56). The biggest spikes are single storms – Hurricane David (1979), the October 2015 flood, Hurricane Matthew (2016) and Debby and Helene (2024) – and big years happen in every decade. The 10-year average has risen since about 2010, so this is worth watching, but with 60 years of data from 10 stations it is not yet distinguishable from natural ups and downs. Total yearly rain, the wettest day and the longest dry spell also show no significant change.

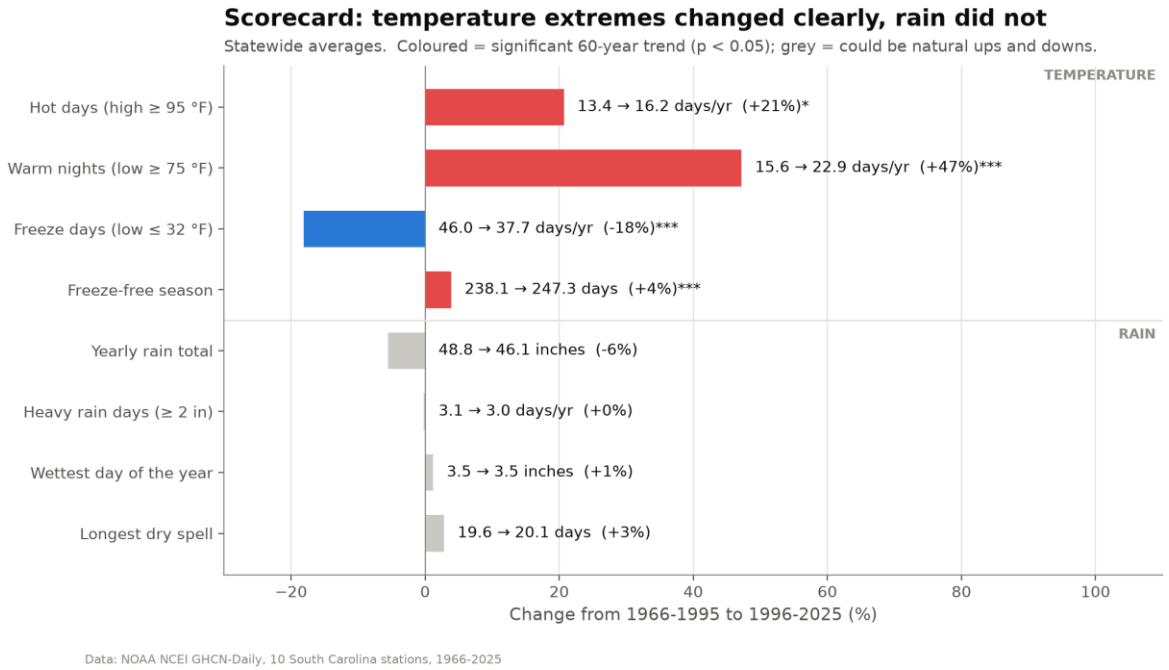


Figure 12. Scorecard of all extreme-weather measures: statewide change from 1966–1995 to 1996–2025. Coloured bars have a significant 60-year trend; grey bars do not.

Table 5. Scorecard of extreme-weather measures (statewide averages).

Measure	1966–1995	1996–2025	Change	Trend p-value
Hot days (high ≥ 95 °F)	13.4 days/yr	16.2 days/yr	+21%	0.046
Warm nights (low ≥ 75 °F)	15.6 days/yr	22.9 days/yr	+47%	< 0.001
Freeze days (low ≤ 32 °F)	46.0 days/yr	37.7 days/yr	-18%	< 0.001
Freeze-free season	238.1 days	247.3 days	+4%	< 0.001
Yearly rain total	48.8 inches	46.1 inches	-6%	0.366
Heavy rain days (≥ 2 in)	3.1 days/yr	3.0 days/yr	0%	0.564
Wettest day of the year	3.5 inches	3.5 inches	+1%	0.531
Longest dry spell	19.6 days	20.1 days	+3%	0.859

Answer to Angle 3: Yes – temperature extremes have clearly shifted. South Carolinians now get many more warm nights, somewhat more hot days, far fewer freezes and a longer growing season. Rain

extremes, however, have **not** changed in a statistically clear way in this data. This contrast is an important result in itself: not every "climate change" claim is supported equally by local data.

8. Task 3 – Visualising the data with Matplotlib

All 12 figures are made with Matplotlib and saved as 200-dpi PNG files in the figures/ folder. Every chart was designed to be read in a few seconds during a presentation:

- **One message per chart.** The title of every figure states the finding (for example "Winter is warming 1.7× faster than the rest of the year"), and a grey subtitle explains how to read it. Titles that contain numbers are generated from the data, so they update automatically if the data changes.
- **The right chart for the job.** Bars for yearly anomalies, ranked horizontal bars for comparing cities, a map for location, a heat-map (warming stripes) for 600 values at once, dumbbell charts for "then vs. now" and "day vs. night", and a scorecard to sum up.
- **Consistent colour meaning.** Coast = blue, Midlands = orange, Upstate = green on every chart; daytime = orange and night-time = purple; red = warmer, blue = cooler. The palette was checked with a colour-blindness validator (all colour pairs pass the deuteranopia/protanopia separation test), and every colour is also backed by a text label.
- **Honest axes and uncertainty.** Bars start at zero, there are no dual y-axes, confidence intervals are drawn as error bars, significance stars are shown, and non-significant results are drawn in grey.
- **Direct labels instead of legends where possible**, a light grid, no chart borders, and a data-source note in the corner of every figure.

All styling lives in one function, utils.set_style(), so every figure looks the same; utils.save() adds the source note and saves the file.

9. Limitations

- **Raw, not homogenised data.** NOAA also publishes adjusted datasets that correct for station moves and instrument changes. I used the raw daily data and a simple neighbour check instead. This caught the obvious problem (Sumter), but smaller jumps at other stations could still be in the data.
- **Urban growth.** Several stations are at airports or in cities that grew a lot since 1966. Part of their warming, especially at night, is local urban heat rather than regional climate (Angle 2 shows this).
- **Simple statistics.** A straight line is a simple model, and its p-value assumes each year is independent. Real climate data has some year-to-year memory (e.g. El Niño), so the true uncertainty is a bit larger than shown. Testing many measures also raises the chance of one false positive; the main findings have p-values far below 0.05, so they are robust to this.
- **Only 10 stations and fixed thresholds.** More stations would give a finer map, and thresholds like 95 °F mean different things in the mountains and at the beach.

10. Conclusion

Using 217,782 days of NOAA weather records from 10 stations, this project found that **South Carolina has warmed by about 3 °F since 1966**, and that the warming is happening at every trusted station in the state. Looking from three angles showed much more than the single average number:

1. **Where:** Columbia warmed fastest (+0.58 °F/decade), statistically tied with Greenville-Spartanburg, Beaufort and Charleston Airport – but every region is warming at a similar rate.
2. **When:** the warming is strongest in **winter** and at **night**, and night-time warming is strongest in the most urban locations.
3. **What people feel:** many more warm nights (+47%), fewer freezes (−18%), a growing season about 3 weeks longer, and somewhat more hot days – but **no clear change in heavy rain yet**.

For a resident, this means milder winters, sticky summer nights that offer less relief, and a longer season for both crops and pests. For a data scientist it is also a reminder to check the data before trusting it – one broken thermometer in Sumter would have changed the answer.

11. How to run the code

The repository is self-contained: the downloaded data is included, so the analysis runs offline.

```
git clone [GitHub repository link]
cd SC-Climate-Shift
pip install -r requirements.txt
python run_all.py              # about 20 seconds
python run_all.py --download   # optional: download fresh data from NOAA first
```

Folder structure:

Table 6. Repository structure.

Path	Contents
run_all.py	Runs every step in order
src/config.py	Stations, years, thresholds, colours
src/download_data.py	Task 1: downloads the NOAA data
src/clean_data.py	Cleaning and yearly measures
src/quality_check.py	Station map and neighbour check (Fig. 1–2)
src/angle1_where.py / angle2_when.py / angle3_extremes.py	The three angles (Fig. 3–12)
src/utils.py	Trend statistics and the shared Matplotlib style
data/raw/	Downloaded NOAA files (trimmed)
figures/, results/	All charts (PNG) and result tables (CSV)
report/, presentation/	This report and the presentation slides

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